

# JINGYUAN LIU

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## RESEARCH INTERESTS

Reward Modeling and Reinforcement Learning for Robotics, World Models for Robotic Manipulation, Continuous and Self-Improving Robot Learning

## EDUCATION

**Fudan University**, Shanghai, China Sep. 2023 – Jun. 2027 (Expected)

*B.S. in Software Engineering* [Transcript] GPA: 3.74/4.00 Average Score: 94/100 Rank: 28/109

**University of Toronto**, Toronto, Canada Sep. 2025 – Dec. 2025

*Exchange Student, Faculty of Applied Science & Engineering* [Transcript] GPA: 4.00/4.00

## PUBLICATIONS & MANUSCRIPTS

- **Jingyuan Liu, Yunpeng Qing, Changxu Wei, Zhi Hou.** “ReAP: From Visual Failure Recognition to Action-Conditioned Risk Prediction for Runtime Failure Monitoring.” Under review at ICLR 2027.
- **ACE-Brain Team.** “ACE-Brain-0.5: A Unified Embodied Foundational Model for Physical Agentic AI.” *arXiv preprint arXiv:2607.04426*, 2026.

## RESEARCH EXPERIENCE

**ACE Robotics, SenseTime**, Shanghai, China Jan. 2026 – Present

*Robotics Algorithm Intern* Advisor: Zhi Hou

- **Action Prediction via Wrist-Camera Motion.** Leading research on predicting robot actions as future wrist-camera poses, using changes in camera position and orientation to represent robot motion.
- **Runtime Failure Monitoring.** Developed **ReAP**, a runtime monitoring framework that combines visual failure detection with action-conditioned future-risk prediction in a failure-relevant latent space.
- **Reward Modeling for Robotic Manipulation.** Built large-scale datasets and benchmarks for task progress estimation and reward modeling, and trained models incorporated into **ACE-Brain-0.5**.
- **Simulation and Real-World Evaluation.** Built policy evaluation pipelines for LIBERO, RoboCasa365, and RoboLab, and tested and debugged policies on AgileX and Galaxea GR1 Pro platforms.

**STARS Laboratory, University of Toronto**, Toronto, Canada Dec. 2025 – Jan. 2026

*Short-Term Project* With Prof. Jonathan Kelly

- Built and validated a Rubik’s Cube dataset for object pose estimation and face-color recognition. [Link]

**Fudan Vision and Learning Laboratory**, Shanghai, China Feb. 2025 – Sep. 2025

*Student Research Intern* Advisor: Zuxuan Wu

- Explored optical flow as an input to Diffusion Policy and compared feature fusion strategies.
- Investigated the generalization of diffusion-based imitation policies to changes in robot initial poses, using training and evaluation trajectories generated with MimicGen.
- Added language conditioning to 3D Diffusion Policy and evaluated multi-task manipulation performance.

**PPIRC, Fudan University**, Shanghai, China Jun. 2024 – Feb. 2025

*Public Performance and Information Research Center*

*Intern* Advisor: Junyu Niu

- Cleaned and processed web data, legal and regulatory documents, and internal audio recordings to support database construction and the development of LLM-based systems for tax-related applications.

## SELECTED COURSE PROJECTS

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- **Robot Navigation System.** Built a ROS-based navigation stack integrating Dijkstra's algorithm and A\* for global path planning with VFH for reactive obstacle avoidance and local navigation.
- **World Model Reproduction and Extension.** Reproduced and extended the World Models framework with VQ-VAE and causal Transformer variants, comparing reconstruction quality, latent prediction accuracy, and control performance on CarRacing and Breakout. [Code]
- **LLM Agent for Inference Optimization.** Built an agent to generate and optimize LLM inference code, using correctness tests and performance feedback to guide iterative GPU kernel optimization.

## HONORS & AWARDS

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|-----------------------------------------------------------|------------|
| Fudan University Outstanding Student Award                | 2024, 2025 |
| Mathematical Contest in Modeling (MCM), Honorable Mention | 2025       |

## ADDITIONAL INFORMATION

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- **English:** IELTS 7.5    TOEFL 101    Duolingo English Test 145
- **Programming:** Python, C++, C, Java